

Size 5 ¼"

Double disc, patent Apple 2x17 pins

Size 5 ¼"

1. DSSI
2. SDI
3. MFM -> ST412
4. SASI
5. ESDI
6. SCSI-I 50
7. IDE
8. MFM (connector\Interface not edge.) -> ST506
9. Lisa patent Apple 2x17 pins

SMD I don't have

ESMD (SMD-E) I don't have

Size 3 ½"

ESDI 72 pins

ESDI 44 pins

ESDI like MFM

MFM -> ST506

IDE\ATA

MFM -> ST412 RLL

JVC unknow

Conner for Apple Hard Disk 40SC 2x17 pins

DSSI

SCSI-I 50

SCSI 68

SCA 80

FC

SSA

SATA gruby

SATA średni fotki wymiary

SATA thin fotki wymiary

SAS

Size ISA 3 ½"

HardCard ISA slot 3 ½"

Size 2 ½"

ESDI

ESDI caddy

SCSI Apple (IBM) 48 pins

IDE

IDE thick

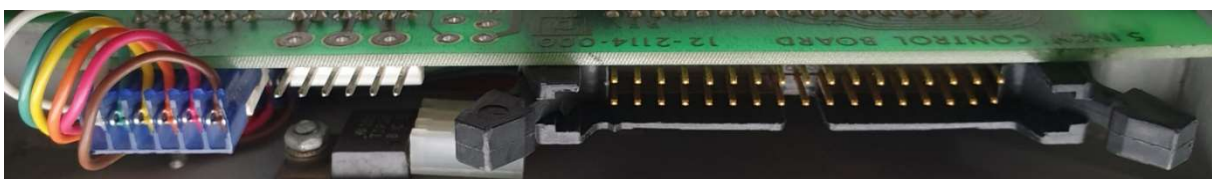
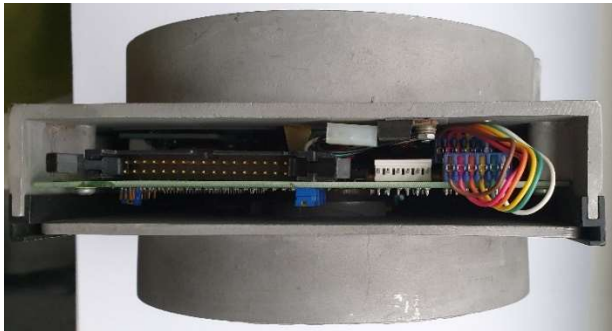
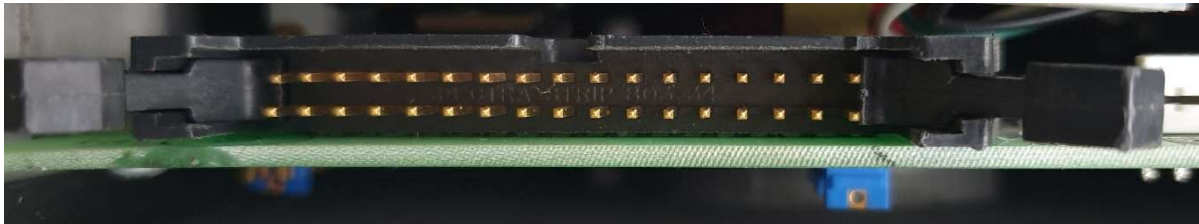
Size 5 $\frac{3}{4}$ " (no $\frac{1}{4}$ ")

5 $\frac{3}{4}$ " Patent Apple 2x17pins ? New World Computer Company INC. Model : Mikro Disc.

This disc consists of two almost identical discs attached to each other at the back.

Connection similar to the proprietary SCSI patent for Apple PowerBook and Lisa.

It has 2 x 17 pins socket. And power socket 6 pins. The situation is identical to the dedicated and proprietary connector in Connor CP-3045 for Apple and Lisa.



Size 5 ¼"

DSSI

DEC Model RF30 5 ¼" DSSI hard drive. 50 pin connector for data. 5 pin power connector. Data Σ=> 50 pins. DSSI **D**igital **S**torage **S**ystems **I**nterconnect. The predecessor of Parallel SCSI, i.e. SCSI-I. A bus for connecting mass storage devices and clustering VAX systems.

<https://gunkies.org/wiki/RF30>



RF30	
	
RF30 with disk rails for BA213/BA215	
Manufacturer:	Digital Equipment Corporation
Year Introduced:	1989
Drive Controller(s):	all DSSI Controllers
Capacity:	150 Mbytes (formatted)
Transfer Rate:	1.5 Mbytes/sec
Throughput:	32 I/Os/sec
Average Access Time:	29 msec
Physical Size:	5.25" half height

SDI

SDI DEC Model RA70 5 ¼" hard drive. SDI Standard Disk Interconnect. Connector 2x10 + 2x4 +2x4 = 36 pin. Power connector 4 pins.
https://gunkies.org/wiki/RA70_disk_drive



RA70	
Manufacturer:	DEC
Drive Controller(s):	all SDI controllers
Capacity:	280 Mbytes (formatted) 350 Mbytes (unformatted)
Transfer Rate:	1.4 Mbytes/sec
Average Access Time:	27 msec
Revolutions per Minute:	4000
1/2 Revolution Time:	7.5 msec
Average Seek Time:	19.5 msec
Total Surfaces:	11 data, 1 servo
Number of Heads:	11 data, 1 servo
Tracks per Surface:	1507
Sectors per Track:	34
Bytes per Sector:	512
Tracks per Inch:	1355
Density:	22,437
Start Time:	59 secs (avg)
Stop Time:	15 secs (avg)
Physical Size:	9cm x 14.6cm x 22.4cm
Weight:	10.5 lbs
Power Consumption:	264 watts (idle)

MFM* Type ST412

Hard drive 5 1/4" MFM*, HH with black face, Seagate ST-225. MFM type: ST412. Two connectors: 34 pins for control and 20 pins for data. $\Sigma \Rightarrow$ 54 pin. Green diode..

<https://stason.org/TULARC/pc/hard-drives-hdd/seagate/ST225-21MB-5-25-HH-MFM-ST412.html>

*)The common name of the interface is MFM. The correct name is ST-506 or ST412. Derived from the disk model. It became the standard. There was no such thing as an "RLL disk or MFM disk". The disk type (MFM or RLL) was determined solely by the low-level format, and this in turn depended on the disk controller card. Theoretically, any ST-506 (two-wire) drive can be formatted using an MFM or RLL controller

MFM - **M**odified **F**requency **M**odulation is the successor FM.

RLL - **R**un **L**ength **L**imited.



Hard Drive: SEAGATE: ST225 21MB 5.25"/HH
MFM ST412

S T 2 2 5 SEAGATE
NO MORE PRODUCED

Native| Translation

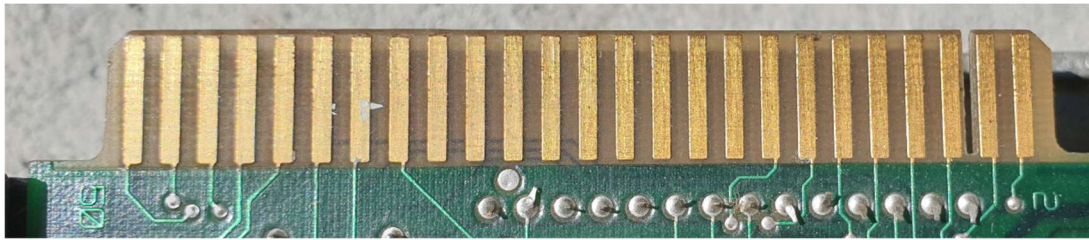
Form	5.25"/HH	Cylinders	615		
Capacity form/unform	21/ 26 MB	Heads	4		
Seek time / track	65.0/20.0 ms	Sector/track	17		
Controller	MFM / ST412	Precompensation	300		
Cache/Buffer	KB	Landing Zone	670		
Data transfer rate	0.625 MB/S int 0.625 MB/S ext	Bytes/Sector	512		
Recording method	MFM	operating		non-operat	ing



SASI

5 ¼" SASI hard drive, Xebec 4020 installed in Amstrad PC1640\PC1520.

Shugart Associates System Interface. Two connectors: 2x23 + 2x2 pins for data and 4 pins for power.
Σ=> 50 pins. Red LED



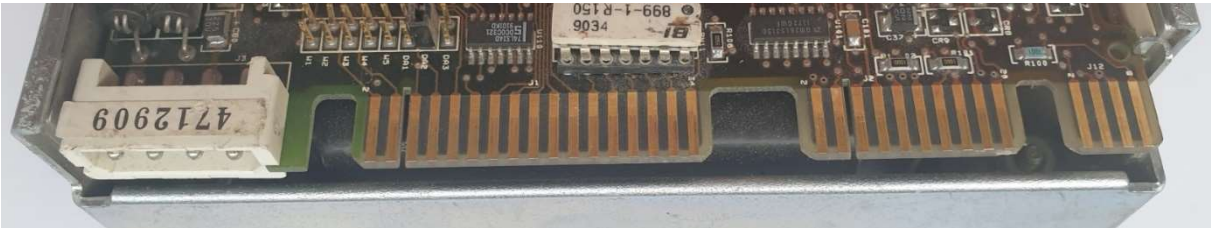
ESDI

5 ¼ Micropoli 1568 Full High
2x17 pins + 2x20 pins + 2x4 pins Σ=> 82 pins Power supply 4 pins.
<https://stason.org/TULARC/pc/hard-drives-hdd/micropolis/1568-15-702MB-5-25-FH-ESDI.html>

"Enhanced Small Device Interface". It was a hard drive interface standard developed in 1983 by Control Data Corporation. ESDI offered several improvements over the earlier ST506 standard, including:

- Higher data throughput: ESDI could transmit data at up to 10 Mbps, while the ST506 was limited to 5 Mbps.
- Larger disk capacity: ESDI supported disks up to 1 GB, while ST506 supported disks up to 640 MB.
- Improved Reliability: ESDI has introduced several features to improve data reliability, such as error correction coding.

ESDI was a popular hard drive interface standard in the 1980s and early 1990s. However, it was replaced by newer standards such as IDE and SCSI, which offered even higher performance and capacity.

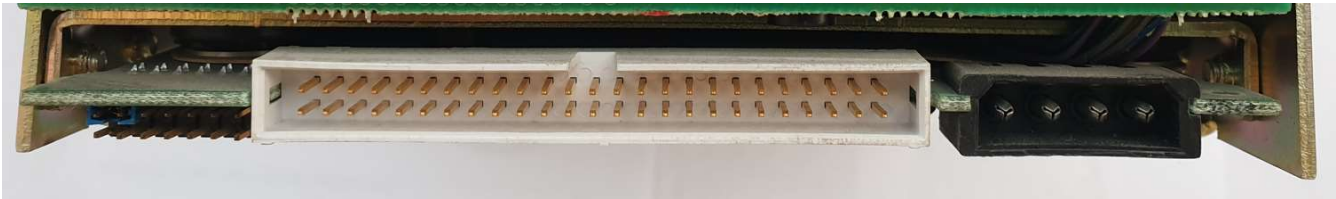


1 5 6 8 - 1 5 MICROPOLIS				
NO MORE PRODUCED		Native Translation		
-----+-----+-----				
Form	5.25"/FH	Cylinders	1632	
Capacity form/unform	702/ 765 MB	Heads	15	
Seek time / track	16.0/ 4.0 ms	Sector/track	28	
Controller	ESDI	Precompensation		
Cache/Buffer	KB	Landing Zone		
Data transfer rate	1.875 MB/S int	Bytes/Sector	1024	
	MB/S ext			
Recording method	RLL	operating		non-operat
ing				

SCSI-I 50 pins

DISK 5 ¼" SCSI 50, FH with black face, Seagate ST4702N. Two cables: 50 pin data and 4 pins for power. Data Σ=> 50 pin. Red LED.

<https://stason.org/TULARC/pc/hard-drives-hdd/seagate/ST4702N-WREN-601MB-5-25-FH-SCSI1-SE.html>



S T 4 7 0 2 N		W R E N	SEAGATE		
NO MORE PRODUCED				Native	Translation
-----+-----+-----					
Form	5.25"/FH		Cylinders	1546	
Capacity form/uniform	601/ 702 MB		Heads	15	
Seek time / track	16.5/ 3.0 ms		Sector/track	50	
Controller	SCSI1 SINGLE-ENDED		Precompensation		
Cache/Buffer	32 KB LOOK-AHEAD		Landing Zone		
Data transfer rate	2.000 MB/S int		Bytes/Sector	512	
	4.700 MB/S ext SYNC				
Recording method	RLL 2/7			operating	non-operat
				ing	
-----+-----+-----					
Supply voltage	5/12 V	Temperature	*C		
Power: sleep	W	Humidity	%		
standby	W	Altitude	km		
idle	W	Shock	g	10	32
seek	21.0 W	Rotation	RPM	3600	
read/write	W	Acoustic	dBa		
spin-up	W	ECC	Bit		
		MTBF	h	100000	
		Warranty Month		12	
Lift/Lock/Park	YES	Certificates			

IDE

Disc 5 ¼" Quantum Bigfoot 4320AT.

Two cables: 40 pin data and 4 pin for power. Data $\Sigma \Rightarrow$ 40 pin. And jumper pins.



MFM (connector\Interface not edge.)

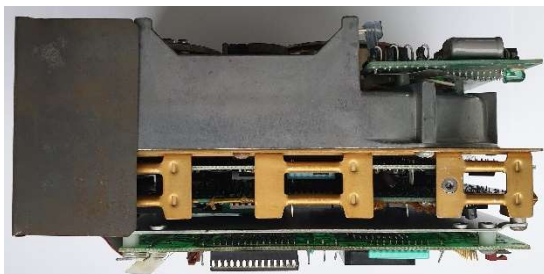
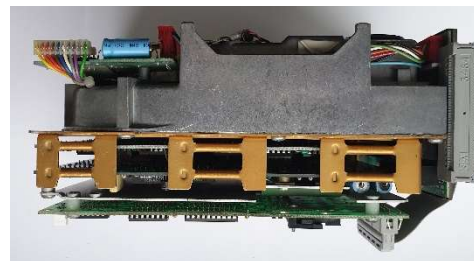
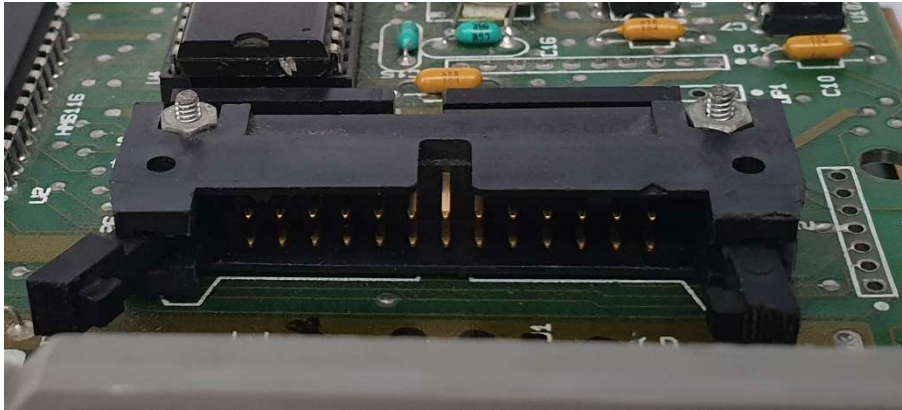
Disc 5 ¼" MFM, HH with black face, Tandon TM 252. MFM type: ST506. Two cables: 2x10 = 20 pin and 2x17 = 34pin $\Sigma \Rightarrow$ 54 pin. Red LED. Mounted, for example, in TECMAR together with Q-PAK. The first external drive for Mac.

<https://stason.org/TULARC/pc/hard-drives-hdd/tandon/TM252-10MB-5-25-HH-MFM-ST506.html>



T M 2 5 2 TANDON		Native Translation	
NO MORE PRODUCED			
Form	5.25"/HH	Cylinders	306
Capacity form/uniform	10/ 13 MB	Heads	4
Seek time / track	85.0/ 3.0 ms	Sector/track	17
Controller	MFM / ST506	Precompensation	128
Cache/Buffer	KB	Landing Zone	305
Data transfer rate	0.625 MB/S int	Bytes/Sector	512
	0.625 MB/S ext		
Recording method	MFM	operating	non-operat
ing			
Supply voltage	5/12 V	Temperature °C	4 50 -40 60
Power: sleep	W	Humidity %	8 80
standby	W	Altitude km	-0.457 3.050 -0.457 12.
195			
idle	W	Shock g	10 40
seek	W	Rotation RPM	3600
read/write	W	Acoustic dBA	
spin-up	W	ECC Bit	
		MTBF h	11000
Lift/Lock/Park	NO	Warranty Month	
		Certificates	

Apple Widget 10 MB 2x17 pins. Patent Apple for Lisa 2 Computer. 1983.
5 ¼" Full High.

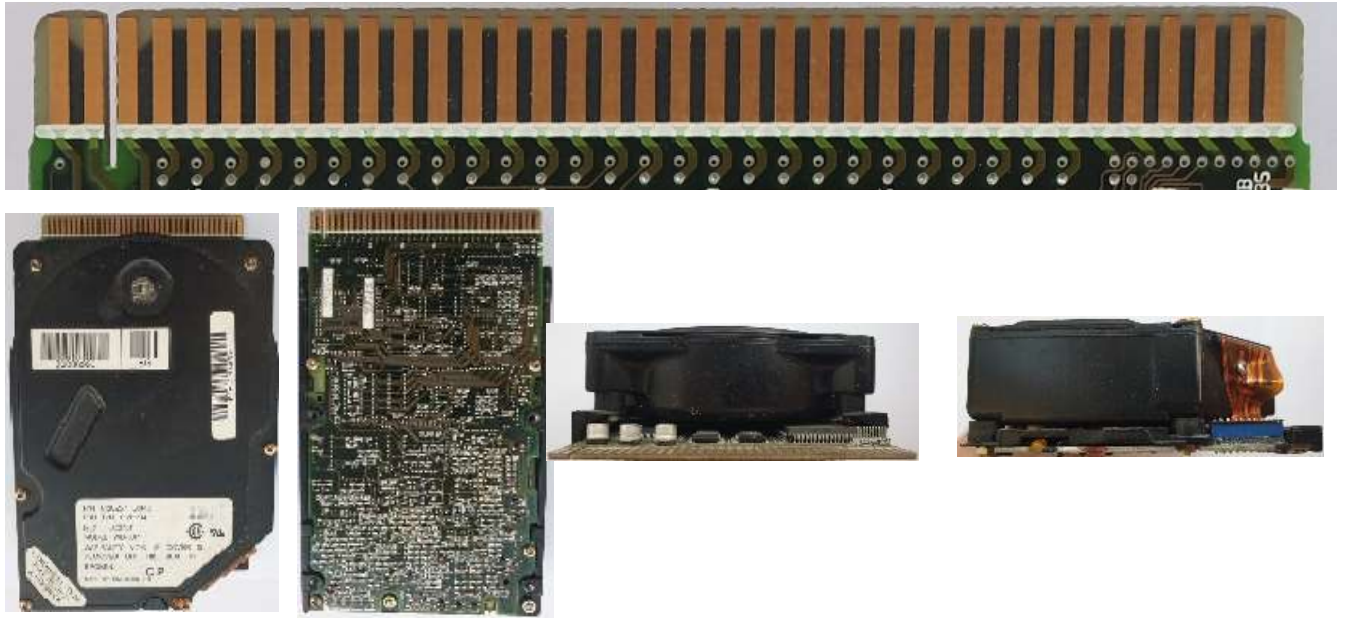


Size 3 ½"

ESDI 72 pins.

3 ½" ESDI HH DISK, HDD from IBM PS. IBM WD-387, PN:6128257 FRU:6128294. 60MB. 2x36 pins.

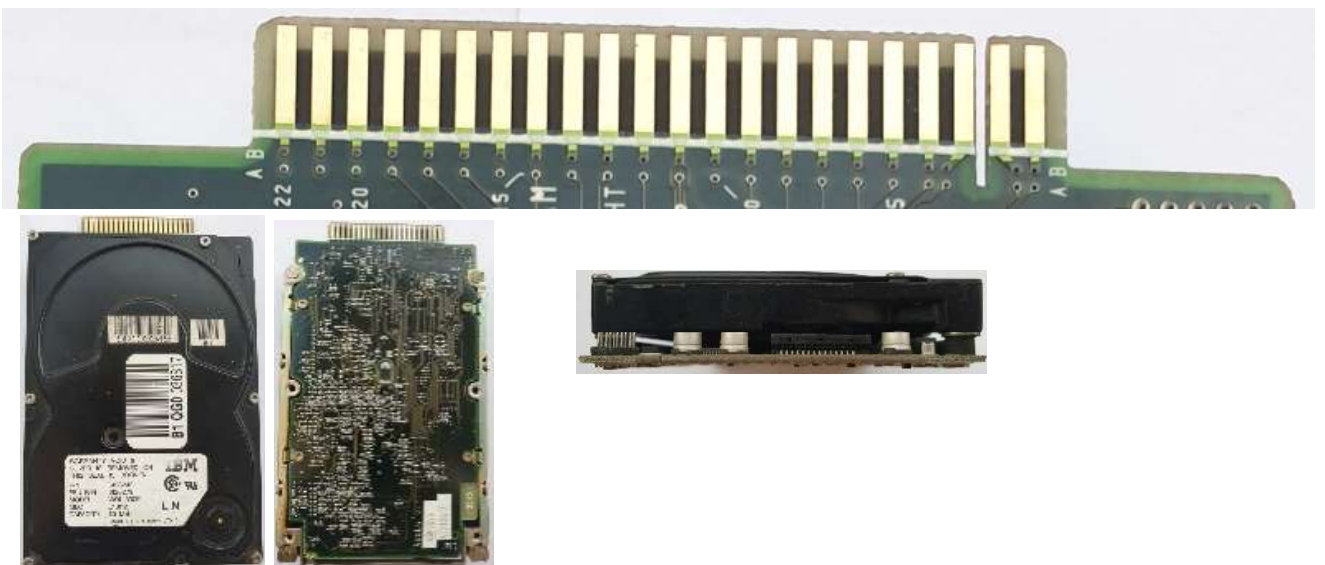
Σ=>72 pins. Yellow LED.



ESDI 44 pins

3 ½" ESDI HDD from IBM PS. WDL-330P, PN:6128242 FRU:6128279. 30MB. 2x22. Σ=> 44 pins.

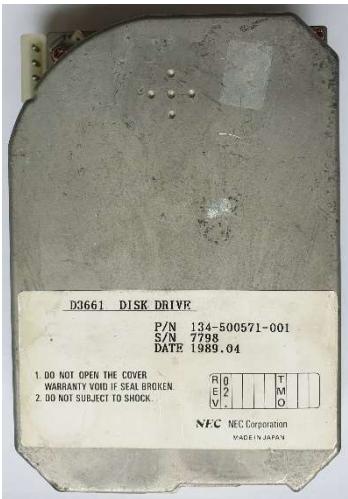
Without diode.



ESDI as MFM

NEC D3661 ESDI HH. Two cables: 2x10 pin + 2x17 pin for power. Separate power supply 4 pins. Σ=> 54 pin.

<https://stason.org/TULARC/pc/hard-drives-hdd/nec/D3661-113MB-3-5-HH-ESDI.html>

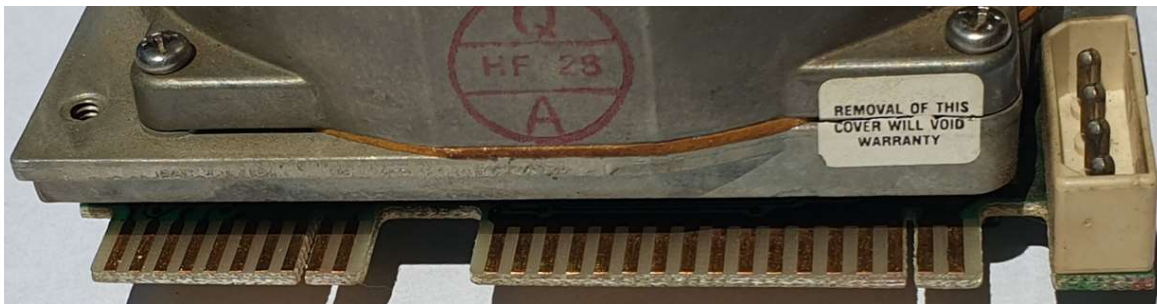


D 3 6 6 1 NEC		Native Translation	
-----+-----+-----+-----			
Form	3.5"/HH	Cylinders	915
Capacity form/uniform	113/ MB	Heads	7
Seek time / track	20.0/ ms	Sector/track	
Controller	ESDI	Precompensation	
Cache/Buffer	KB	Landing Zone	
Data transfer rate	1.250 MB/S int MB/S ext	Bytes/Sector	512
Recording method	RLL 2/7	operating	non-operating
ing			
-----+-----+-----+-----			
Supply voltage	5/12 V	Temperature	*C
Power: sleep	W	Humidity	%
standby	W	Altitude	km
idle	W	Shock	g
seek	W	Rotation	RPM
read/write	W	Acoustic	dBA
spin-up	W	ECC	Bit
		MTBF	h 30000
		Warranty Month	
Lift/Lock/Park	YES	Certificates	

MFM Type ST506

MFM HH Kyocera KC-20B. 21MB. MFM type: ST506. Two cables: 2x10 pin + 2x17 pin for power. Separate power supply 4 pins. $\Sigma \Rightarrow$ 54 pin. Green diode.

<https://stason.org/TULARC/pc/hard-drives-hdd/kyocera/KC-20B-21MB-3-5-HH-MFM-ST506.html>



K C 2 0 B KYOCERA
NO MORE PRODUCED

Native | Translation

Form	3.5"/HH	Cylinders	615			
Capacity form/uniform	21/ 36 MB	Heads	4			
Seek time / track	62.0/15.0 ms	Sector/track	17			
Controller	MFM / ST506	Precompensation	300			
Cache/Buffer	KB	Landing Zone	615			
Data transfer rate	0.625 MB/S int 0.625 MB/S ext	Bytes/Sector	512			
Recording method	MFM	operating non-operat ing				

ATA/IDE.

3 1/2" ATA/IDE DISK. 40 pins + 9 configuration jumpers. Separate power supply 4 pins. $\Sigma \Rightarrow$ 40+9 pins.



MFM Type ST412 with **RLL**

3 1/2" disc ST-157R Seagate RLL* - Run Length Limited. Type ST412. Two cables: 2x10 pins + 2x17 pin for power. Separate power supply 4 pins. Σ=> 54 pins.

<https://stason.org/TULARC/pc/hard-drives-hdd/seagate/ST157R-0-49MB-3-5-HH-RLL-ST412.html>



ST157R-0 SEAGATE
NO MORE PRODUCED

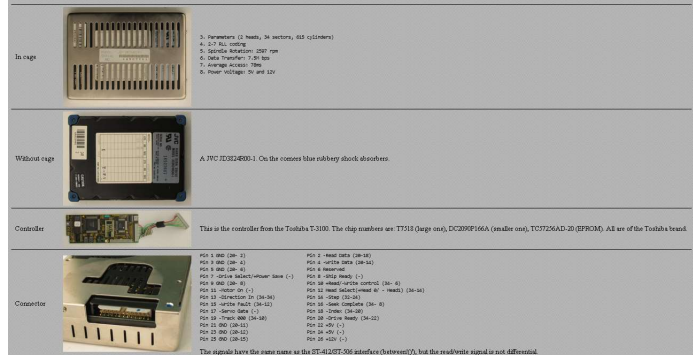
Native | Translation

Form	3.5"/HH	Cylinders	615		
Capacity form/uniform	49/ 58 MB	Heads	6		
Seek time / track	38.0/ 8.0 ms	Sector/track	26		
Controller	RLL / ST412	Precompensation	65535		
Cache/Buffer	KB	Landing Zone			
Data transfer rate	0.938 MB/S int	Bytes/Sector	512		
	0.938 MB/S ext				
Recording method	RLL 2/7	operating		non-operat	
ing					

3 1/2" disc. **Standard - unknown.** JVC JD-3824R00-1. Σ=>26 pins. There are no equivalents \ substitutes \ substitutes. Mounted in Epson PX-16, Epson LT, Epson Q150A, Sharp PC-7200.



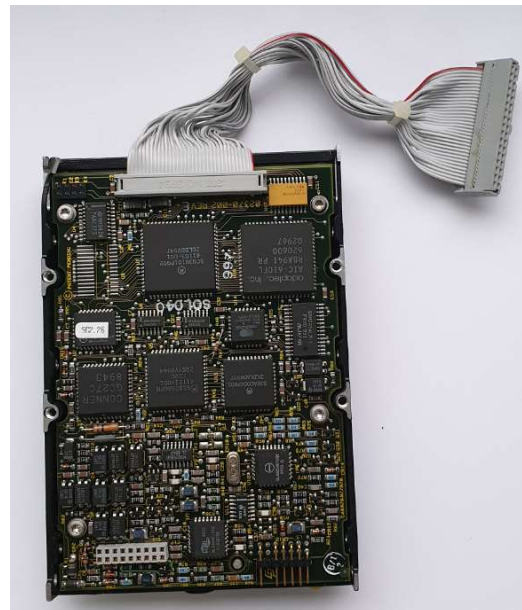
- Epson FX-16
- Epson Equity LT
- Epson Portable PC (Q150A)
- Sharp PC-7200
- Data General One Model ZT ?
- GfKDCase 3 Phase
- Toshiba T1200 ?
- Wang WLTC Laptop



SCSI - patent for Apple 2x17 pins.

Conner CP3045 (Apple Hard disk 40SC)

Only 2x17 pins = 34 pins. The connector and number of pins are the same as in the Apple Lisa disk and as in the "strange" disk 5 ¾". A drive designed specifically for Apple PowerBook. Known by Apple as "Hard Disk 40SC". It has a patented SCSI connector. The cable is permanently connected to the disk. There is no power socket. Only 2x17 pins cable. SCSI patent for Apple.

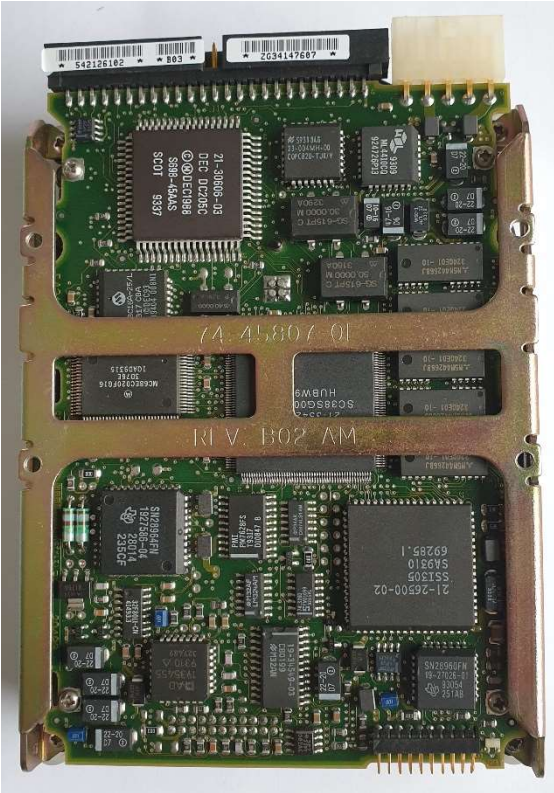


DSSI

DEC Model RF35 3½" Data 2x25=50 pins. Power 5 pins. Data Σ=> 50 pin.

DSSI Digital Storage Systems Interconnect.

<https://gunkies.org/wiki/RF35>



RF35	
Manufacturer:	Digital Equipment Corporation
Drive Controller(s):	all DSSI Controllers
Capacity:	852 Mbytes (formatted)
Transfer Rate:	4.0 Mbytes/sec
Average Access Time:	15.1 msec
Average Seek Time:	9.5 msec
Start Time:	< 30 sec
Stop Time:	< 15 sec
Physical Size:	3.5" half height

SCSI-I 50 pins.

Quantum Fireball ST. 2x25 pins. Power 4 pins. $\Sigma \Rightarrow$ 50 pins.



SCSI 68 pins.

3 1/2" SCSI 68 Quantum Viking disc. 2x20 pins. 2x6 pins configuration jumpers. Power supply separately 4 pins. $\Sigma \Rightarrow$ 40 + 12 pins.



SCA - UltraSCSI 3 80 pins. Single Connector Attachment.

3 ½" SCA drive = Ultra 3 SCSI 80 IBM Ultrastar. 2x40 pins. $\Sigma \Rightarrow$ 80 pins.



FC Fibre Channel.

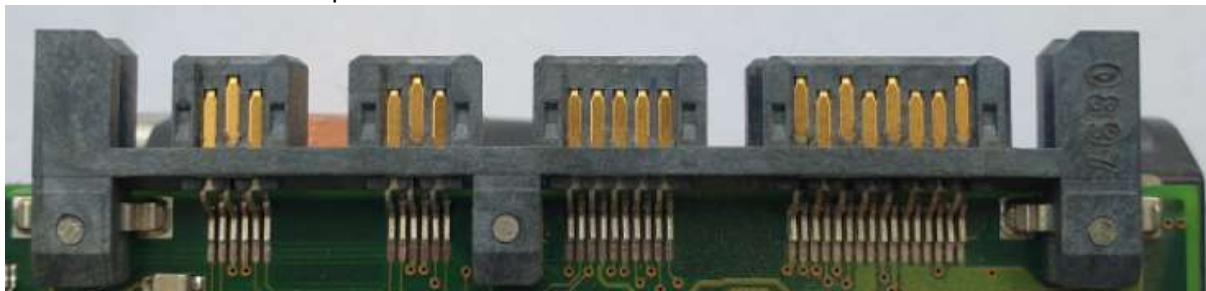
Seagate Cheetah. Pins 2x20 $\Sigma \Rightarrow$ 40 pin.



SSA (Serial Storage Architecture)

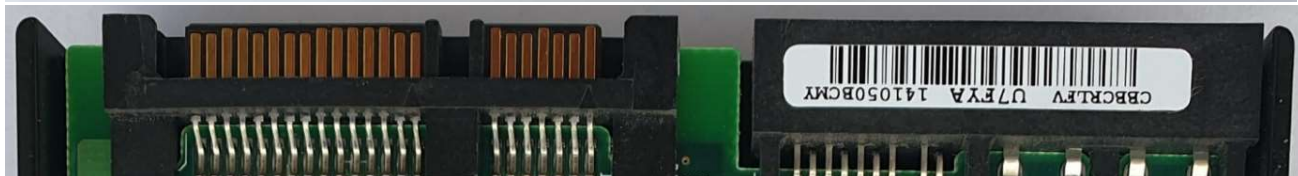
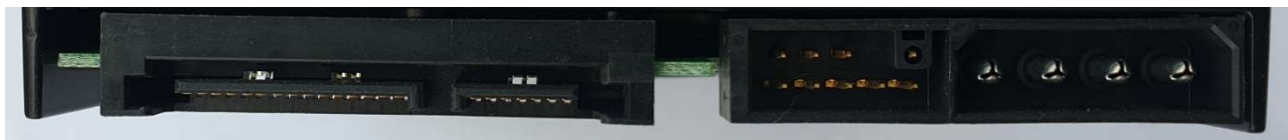
Drive 3 1/2" Pins : $2 \times 3 + 2 \times 3 + 2 \times 5 + 2 \times 8 = \Sigma \Rightarrow 38$ pins.

and on the other side 2×16 pins.



SATA Serial ATA.

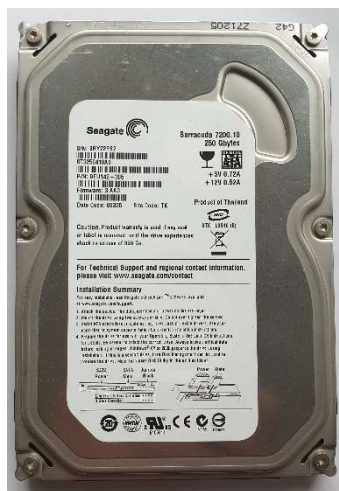
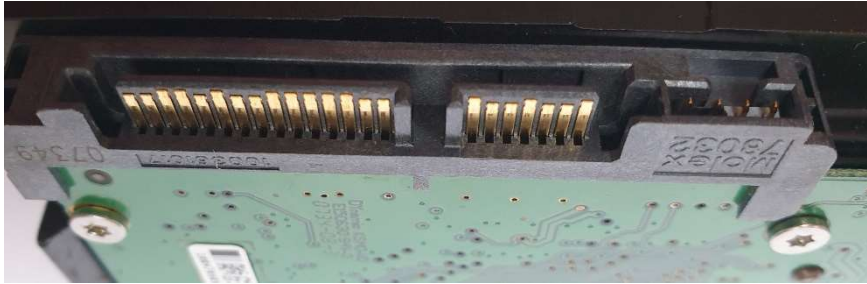
Drive 3 1/2" Pins : 15 + 7 + 9 jumpers + 4 pins for power.



SATA Serial ATA.

Drive 3 1/2" Pins : 15 + 7 + 9 jumpers + 4 pins for power.

Low profile. Thickness = 2 cm.



SAS Serial **A**ttached **S**torage.

Drive 3 ½" Pins : 15 + 7 + 4 jumpers.





- MFM ST-412
- ESDI IBM PS/2
- MFM IBM ST-506
- IDE \ EIDE
- SCSI 50
- SCSI 68
- SCSI 80
- SSA
- FC
- SATA
- SAS



- MFM ST-412
- ESDI IBM PS/2
- MFM IBM ST-506
- IDE/ATA
- SCSI 50
- SCSI 68
- SCSI 80
- SSA
- FC
- SATA
- SAS

Size 3 ½" ISA

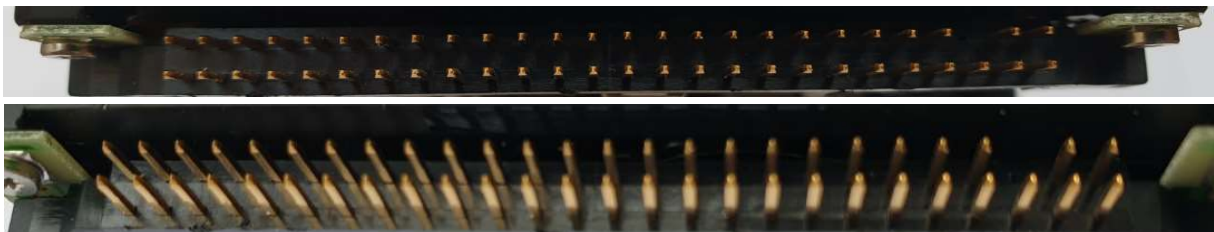
Drive 3 ½" ISA Slot, HardCard 20 PLUS. That is, the disk on the ISA card. Industry Standard Architecture.



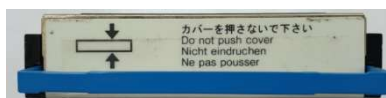
Size 2 ½"

ESDI Enhanced Small Disk Interface.

Drive 2 ½" IBM WD-W2160D, **h=17mm**, 2x24 pins + 2x2 pins = 48+4 $\Sigma \Rightarrow$ 52 pins.

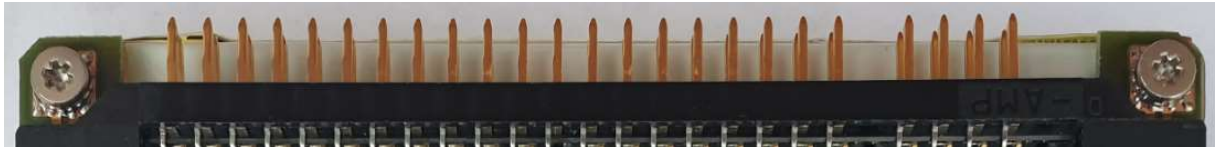


ESDI - caddy IBM WD-W2160D, **h=18mm**, case\caddy for the above-mentioned disk. 2x30 pins $\Sigma \Rightarrow$ 60 pins.



SCSI Apple (IBM)

Drive 2 ½" WDS-280 h=17mm, 2x20 pins + 2x4 pins $\Sigma \Rightarrow$ 48 pins.



IDE thick

TOSHIBA MK2720FC h=18.5mm, 2x22 pins + 2x2 pins $\Sigma \Rightarrow$ 48 pins.



IDE

H=9.5mm. 2x22 pins + 2x2 pins $\Sigma \Rightarrow$ 48 pins.



IDE. Half length. Thickness h=9.5mm. 2x22 pins + 2x2 pins $\Sigma \Rightarrow$ 48 pins.



SCSI U320. ST936701LC. h=14.5mm. 2x40 pins. $\Sigma \Rightarrow 80$ pins.



SATA. Hitachi. Thickness h=7mm. Pins 15+7. $\Sigma \Rightarrow 22$ pins.



SATA. SSHD. Hitachi. Thickness $h=7\text{mm}$. Pins $15+7$. $\Sigma \Rightarrow 22$ pins.



SATA. SSD. Intel. Thickness $h=7\text{mm}$. Pins $15+7$. $\Sigma \Rightarrow 22$ pins.



SATA. SSSE Samsung Solid State Enterprise - **Hybrid**. Intel. Thickness h=9.5mm.

Pins 15+7. $\Sigma \Rightarrow 22$ pins.



SATA. 8MB buffer. Toshiba. Thickness h=9.5mm. Pins 15+7. $\Sigma \Rightarrow 22$ pins.





SAS. Sun\Fujitsu. Thickness h=14.5mm. Pins 15+7. $\Sigma \Rightarrow 22$ pins.



USB. Western Digital. Thickness h=14.5mm. Pins 5+4+6. $\Sigma \Rightarrow 15$ pins.
White LED.



SATA. RDX IBM.





<- caddy do n\w HDD ESDI

<- ESDI

<- SCSI

<- IDE *Toshiba GRUBY*

<- IDE „zwykły”

<- SCSI U320

<- SATA

<- SAS

<- USB

SIZE 1.8"

HDD 1.8" PCMCIA CF+ Typ II 340MB 3600 rpm IBM. PN: 22L0046, Pin 2x25. Σ=> 50 pins.



HDD 1.8" PCMCIA iOmega Klik! ZIP 40MB and HDD microdrive PCMCIA Typ I.



PCMCIA it is short from Personal Computer Memory Card International Association

Type I: 16-bit. Configuration thickness: 3.3 mm (0.13 in).

Type II: 16-bit or 32-bit. Configuration thickness: 5.0 mm (0.20 in).

Type III: 16-bit or 32-bit. Configuration thickness: 10.5 mm (0.41 in).

Type IV: 16 millimeters (0.63 in).

HDD 1.8" PCMCIA IBM 2x34. $\Sigma \Rightarrow$ 68 pins. PCMCIA Typ IV.

Height 16 mm.



HDD 1.8" Toshiba PATA 2x25. $\Sigma \Rightarrow$ 50 pins.

Height 8 mm



HDD 1.8" Toshiba mSATA 7+7+2. $\Sigma \Rightarrow$ 16 pins.



HDD 1.8" Apacer SLC HALF (half the length) Slim SATA 15+7. $\Sigma \Rightarrow$ 22 pins.



Height 8 mm.



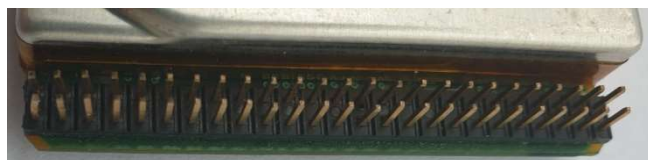
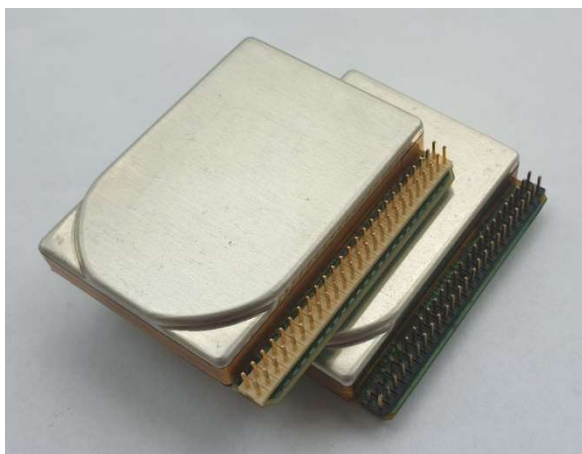
SIZE 1.3"

HDD 1.7" KityHawk HP\Citizen, IDE. Pins 2x25 $\Sigma \Rightarrow$ 50 pins.

Introduced by HP, manufactured by Citizen Corporation.

C3013A 20MB to HP Kittyhawk I.

C3014A 40MB to HP Kittyhawk II.



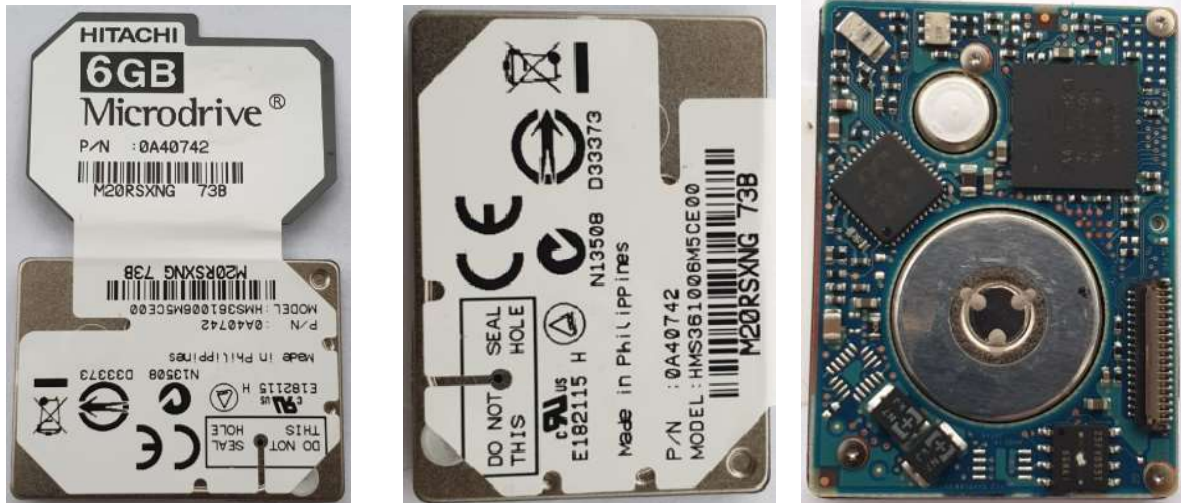
C3013A 20MB to HP Kittyhawk I



C3014A 40MB to HP Kittyhawk II.

SIZE 1"

HDD 1" Hitachi.



SIZE 0.85"

HDD 0.85" Toshiba from Nokia N91.





Różnice między interfejsami ST412 i ST506:

Szybkość transferu danych:

- **ST412:** 5 Mb/s
- **ST506:** 10 Mb/s

Kodowanie danych:

- **ST412:** MFM (Modified Frequency Modulation)
- **ST506:** MFM (Modified Frequency Modulation)

Pojemność dysku:

- **ST412:** Maksymalnie 20 MB
- **ST506:** Maksymalnie 40 MB

Złącze:

- **ST412:** 20-pinowe
- **ST506:** 34-pinowe

Złącza w interfejsach ST412 i ST506 służą zarówno do sterowania, jak i do transferu danych.

Złącze 20-pinowe w interfejsie ST412:

- **Pin 1-10:** Transfer danych
- **Pin 11-16:** Sterowanie
- **Pin 17-20:** Zasilanie

Złącze 34-pinowe w interfejsie ST506:

- **Pin 1-20:** Transfer danych
- **Pin 21-30:** Sterowanie
- **Pin 31-34:** Zasilanie

Dodatkowe piny w złączu ST506 umożliwiają przesyłanie większej ilości danych i zapewniają lepszą kontrolę nad dyskiem twardym.

Dyski IBM WDL-330P używają interfejsu ESDI (Enhanced Small Device Interface). Interfejs ESDI jest następcą interfejsu ST506 i oferuje kilka kluczowych zalet:

- **Wyższa przepustowość:** ESDI może przysyłać dane z prędkością do 10 Mb/s, podczas gdy ST506 jest ograniczony do 5 Mb/s.

- **Większa pojemność dysku:** ESDI obsługuje dyski twarde o pojemności do 1 GB, podczas gdy ST506 obsługuje dyski o pojemności do 40 MB.
- **Ulepszone zarządzanie błędami:** ESDI posiada wbudowane funkcje sprawdzania i korekcji błędów, co zapewnia większą niezawodność danych.

Interfejs MFM nie jest obsługiwany przez dyski IBM WDL-330P.

Dodatkowe informacje:

- Dyski IBM WDL-330P to dyski twarde o wysokości 5,25 cala i pojemności 330 MB.
- Dyski te były popularne w komputerach klasy PC w latach 80.